

INC-2026-0805 - Double Size Conversion in Dropship Feed Importer

Incident Report | Engineering

Document ID	INC-2026-0805	Seller ID	S004
Department	Engineering	Classification	CONFIDENTIAL
Owner	Vikram Singh	Created Date	2026-08-22
Related Scenario ID	SC-006	Document Type	Incident Report

1. Incident Summary

Field	Value
Incident ID	INC-2026-0805
Severity	SEV-2
Service	Catalog Service - dropship product feed importer
Component	Size attribute normalisation
Introduced	2026-07-14
Detected	2026-08-05 12:15 IST
Mitigated	2026-08-12
Resolved	2026-08-19
Incident owner	Vikram Singh, Engineering
Seller affected	S004 - StrideOne Footwear
Related ticket	TKT-2026-0805-301

A schema update to the dropship feed importer caused the UK-to-EU footwear size conversion to be applied twice for any partner supplying UK sizing. Published size charts were approximately two sizes smaller than the physical product. The defect ran in production for 36 days and drove the seller's return rate from a 9 percent baseline to 27 percent.

2. Impact

Dimension	Detail
Time in production	36 days, 14 July to 19 August
SKUs affected	approximately 13,400 of 18,000
Returns in the affected window	2,860
Returns above baseline	approximately 1,950
Peak return rate	27 percent, against a 9 percent baseline
Customers offered a free exchange	2,860
Exchanges accepted	1,410

Dimension	Detail
Reverse logistics cost absorbed	INR 8,70,000
Sellers affected	1 (S004)

The impact did not end with the fix. The elevated return rate remained in the trailing 30-day average after correction and triggered an automated account suspension on 8 September 2026, three weeks after the defect was resolved. That suspension is recorded separately as scenario SC-012 and decision DR-2026-047.

This is the longest-lived defect in the dataset and the only one whose consequences propagated into a commercial risk determination.

3. Detection

Detected by the seller, 22 days after introduction.

StrideOne's marketplace lead raised a ticket on 5 August reporting a return rate rise from 9 to 27 percent over three weeks, with 'size too small' having grown from 14 percent to 71 percent of return reasons, across all four logistics partners and all product lines, with no change on the seller's side.

Support sampled twenty SKUs across four brands within hours and found that brands supplying UK sizing were published approximately two sizes small while a brand supplying EU sizing was correct. That sampling produced the double-conversion hypothesis before the ticket reached Engineering.

No platform control detected this. There was no validation comparing published size distribution against source, and no alert on a return reason code shifting from 14 percent to 71 percent of volume.

4. Timeline

Date	Event
2026-07-14	Feed importer schema update deployed; defect introduced
2026-07-15 to 07-18	Affected SKUs republished through normal feed cycles
2026-07-19	Return rate 11 percent; within normal variation
2026-07-26	Return rate 17 percent; not yet investigated
2026-08-02	Return rate 24 percent; seller begins analysis
2026-08-05	Seller raises ticket with reason-code evidence
2026-08-05	Support sampling confirms double conversion pattern
2026-08-05	Escalated to Engineering
2026-08-06	Defect located in the size normalisation path
2026-08-07	Raised to P1 on commercial impact
2026-08-12	Conversion logic corrected and deployed
2026-08-19	Full catalogue republished with corrected size charts
2026-08-19	Structural feed validation deployed
2026-08-21	Seller confirms sampling correct; ticket closed

5. Symptoms

- Return rate rising steadily over three weeks rather than spiking.
- 'Size too small' dominating return reason codes at 71 percent.
- Pattern present across all four logistics partners and all product lines.
- Brands supplying EU sizing unaffected.
- No import errors, rejections or warnings in the feed pipeline.
- Individual product records looked entirely plausible on inspection.

The last symptom is the heart of this incident. A published EU 40 for a shoe that is actually EU 42 is not detectably wrong from the record alone. It is only wrong relative to the source, and only visible as a systematic shift across the population.

6. Affected Services

Service	Role	State
Dropship feed importer	Normalised partner attributes	Defective: conversion applied twice
Size conversion table	Mapped UK, EU and US conventions	Correct; used twice
Catalog Service	Published listings and size charts	Publishing defective data faithfully
Returns pipeline	Captured return reason codes	Correct; the only source of signal
Risk rules engine	Consumed return rate	Correct; acted on genuine data

7. Investigation

7.1 The Defect

Before 14 July, size normalisation was a single step: read the partner's declared size and its convention, then convert to the published EU convention.

The 14 July schema update introduced a `normalised_size` field to the partner staging record, intended to hold the converted value so that downstream steps would not need the conversion table. The update added the conversion at the staging write step.

The publication step was not changed. It continued to read the size field and apply the conversion before publishing.

After the update, the staging record already held a converted value, and the publication step converted it again. A UK 8 became EU 42 at staging, then EU 42 was treated as a UK value and converted again to approximately EU 47 in internal units, which after the chart mapping presented to buyers as a shoe roughly two sizes smaller than reality for the size label shown.

Partners supplying EU sizing were unaffected, because the conversion is a no-op for values already in the target convention.

7.2 Why Testing Did Not Catch It

The change was tested with unit tests on the new staging write step, which correctly produced a converted value, and with an end-to-end import test using a fixture file. The fixture used EU sizing, because it had been created from a partner sample that happened to be EU-based.

The end-to-end test therefore exercised the only path where double conversion is invisible.

7.3 Why This Class Of Defect Is Hard

The onboarding project for this account recorded a design note in August 2025 that turned out to be exactly right:

"Size conversion is the only transformation in this feed that cannot be validated by inspecting a single record. A wrong colour is obvious; a wrong size is only detectable by comparing the published chart against the physical product or against return data. Validation must therefore be structural rather than per-record."

The structural validation that note called for was recorded as open item OI-2 with a target of January 2026. It was not implemented. It was delivered on 19 August 2026, seven months late and five days after this defect was confirmed.

8. Root Cause

A schema update to the dropship feed importer on 14 July 2026 added size conversion at the staging write step without removing the existing conversion at the publication step. Sizes supplied in UK convention were therefore converted twice, publishing size charts approximately two sizes smaller than the physical product for approximately 13,400 SKUs.

9. Contributing Factors

1. **Conversion added without removing the existing conversion**, a transformation applied at two layers.
2. **Test fixture used EU sizing**, the one convention for which the defect is invisible.
3. **No structural validation** of published size distribution against source, despite a design note requiring it and an open item tracking it.
4. **Open item OI-2 carried for seven months** without escalation.
5. **No alert on return reason code distribution**, the only available signal.
6. **The seller cannot inspect goods**. In a dropship model no one in the chain physically checks the product against its listing.

10. Mitigation

There was no fast mitigation. Correcting the conversion on 12 August fixed future imports, but the incorrect size charts remained published until the catalogue was fully republished on 19 August.

A rollback of the 14 July schema update was considered and rejected. The update had other dependencies that had accumulated over four weeks, and rolling back would have reintroduced a different set of defects. Correcting forward was the lower-risk path.

In-flight returns continued for a further two weeks after republication, because customers who had already received wrongly sized goods returned them on their own timescale. The return rate therefore lagged the fix by approximately three weeks, which is precisely what later triggered the automated suspension.

11. Resolution

Action	Completed
Conversion removed from the publication step	2026-08-12
Full catalogue republished with corrected size charts	2026-08-19
Structural feed validation deployed	2026-08-19

Action	Completed
Test fixtures extended to cover UK, EU and US sizing	2026-08-14
Free exchanges offered to 2,860 affected customers	2026-09-05
Reverse logistics cost absorbed by the platform	2026-09-15

11.1 Structural Validation

The validation compares the distribution of published sizes against the distribution in the partner source. A double conversion shifts the entire distribution while leaving every individual record plausible, so a distribution comparison detects it where record-level checks cannot.

The rule rejects a feed whose published size distribution differs from source by more than a configured tolerance, and routes it for operator review.

12. Validation

Check	Method	Result
Conversion correct	200 SKUs across all four partners sampled against source	All correct
Structural rule detects the defect	Rule back-tested against the 14 July feed	Import rejected on the day of introduction
No false rejections	Rule run against 12 weeks of historical feeds	Zero rejections on clean feeds
Test coverage	Fixtures for UK, EU and US sizing	All three conventions exercised end to end
Return rate recovery	Weekly monitoring	27 percent peak to 10 percent by 27 September

The back-test result is the most useful figure in this report: the control that now exists would have rejected the defective import on the day it was introduced, preventing 36 days of exposure, 1,950 excess returns and an account suspension.

13. Preventive Actions

ID	Action	Owner	Target	Status
PA-1	Remove duplicate conversion from the publication step	Vikram Singh	2026-08-12	Complete
PA-2	Structural size distribution validation	Vikram Singh	2026-08-19	Complete
PA-3	Test fixtures covering all size conventions	Vikram Singh	2026-08-14	Complete
PA-4	Alert on return reason code distribution shift	Neha Gupta	2026-10-16	In progress

ID	Action	Owner	Target	Status
PA-5	Open items past target date escalated automatically	Neha Gupta	2026-09-30	Complete
PA-6	Partner feed contract review to remove conversion entirely	Neha Gupta	2026-11-13	Not started
PA-7	Advance notice to seller before size attribute schema changes	Neha Gupta	2026-11-13	Not started

14. Follow-Up

Two follow-ups extend beyond this incident.

The first is SC-012. The automated suspension on 8 September was triggered by a trailing 30-day return rate that was still elevated by this defect. The risk rule is being recalibrated to exclude return volume attributable to a recorded platform defect and to incorporate a trend component.

The second is PA-6. The conversion exists only because partners supply sizes in three conventions. Requiring a single convention in the 2027 partner contracts would remove the transformation, and therefore this entire class of defect, rather than validating around it.

Post-incident review held 22 August 2026. The review's principal finding was not the coding error but the seven-month-old open item: the control that would have prevented this was identified at onboarding, agreed, tracked, and never delivered.